

the above claim was refuted by Blu-ray, which said that BD-ROMs will be 50 GB at launch, although no launch date was mentioned. Doherty had this to say about the 50 GB launch: "The 50 GB claim for BD-ROM discs is unproven and will not be available for many years to come, based on discussions with major Japanese and US replicators. Replicators not only do not have test lines running, they cannot even pre-order the equipment to begin evaluating this disc. They cannot judge the cost of these discs, or even whether they can be manufactured at all. Major replicators can mass-manufacture 30 GB HD-DVD discs today..."

There are more computer manufacturers backing Blu-ray, again because of the storage capacity aspect. Blu-ray is ready to go with 50 GB, and the theoretical limit for a four-layer HD-DVD disc is only 60 GB. The present might be in favour of HD-DVD, but remember that a 100 GB BD has already been developed!

In terms of the security and anti-piracy features that will be offered by the two formats, Blu-ray has BD+ and ROM-Mark, which HD-DVD doesn't offer (see box *Enhanced Security In Blu-ray*). That means in theory, Blu-ray discs will be tougher on pirates than HD-DVDs, which is good news for the studios.

News of hardware support for Blu-ray keeps coming in. Just last month, it was reported that Panasonic would soon begin sampling what it claims is "the first optical drive control chipset capable of writing to any recordable or re-writable disc format"—except HD DVD. Not much of a surprise, since Panasonic's parent company Matsushita is backing Blu-ray. The chipset specifications list format support, and amongst these are CD-ROM, CD-R/RW, DVD-ROM, DVD-RAM, DVD±R/RW, BD-ROM, and BD-R/RE. No HD-DVD here, definitely! Panasonic said mass production of the chipset would begin in January next year.

One could look to history and see that this is just another format war. Case in point: when VCRs were introduced, there were two competing formats. Sony was pushing Betamax, considered technologically superior; JVC and Matsushita were pushing VHS. Sony refused to

When VCRs were introduced, Sony was pushing Beta-max, considered technologically superior; JVC and Matsushita were pushing VHS. VHS became the standard. Sony is now pushing the technologically superior Blu-ray

license Betamax to OEMs, while JVC and Matsushita licensed VHS to whoever wanted it. VHS became the standard. Sony is now pushing the technologically superior Blu-ray...

Talking about Betamax, one of the reasons your old video cassettes are VHS and not Betamax is because pornographers took to VHS and not Betamax. Like it or not, much of entertainment technology has been driven by "The Other Hollywood." The porn industry releases over 11,000 new titles on DVD every year, compared to about 400 by Hollywood. In which case, we don't know anything right now about whether HD-DVD or Blu-ray will win—the fate of the format wars might lie in the hands of the pornographers.

It may not even be Blu-ray or HD-DVD at all: it may be EVD, China's next-generation DVD format! EVD is a red-laser format, and promises capacities of 16 GB. So what's so great about EVD? Not much, except that China wants it. That country produces 70 to 80 per cent of the world's DVD players, and manufacturers pay a large amount in terms of licensing fees to the international owners of the intellectual property patents for DVD players. China is not happy.

According to the state-owned news agency Xinhua, China will put EVD on the market "before 2008." Hollywood is behind much of the development that's happening in the high-definition storage arena, but what about China's thriving movie industry, and Bollywood as well—considering that China's biggest potential export market is India? Things could change as 2008 comes by.

There is some weight behind the idea that it will be the consumers who will ultimately decide what format will rule. So if HD-DVD is initially cheaper, which it might well be, it could gain the advantage—and take over from there.

This opinion is, in fact, shared by a large number of people: that they would upgrade their DVD players when the alternative became cheap enough—so it makes sense to think that whichever format becomes "cheap enough" first will obviously be "the winner."

It could even be something as silly as a name! Historically, consumers have been more likely to adopt a new product when it comes as a natural progression from what it is replacing. HD-DVD seems the natural progression of DVD, in terms of the name. "HD-DVD" sounds like something to do with DVD, and, admit it, "Blu-ray" sounds like the name of a dangerous fish!

In Conclusion

The blue-laser DVD format war has been on for some time now, and will be on for the foreseeable future. There are far too many aspects to it to have explored here, and we can only hope you've gained a foothold on what's going on in this space. 2006 will be an exciting year in the unfolding of this drama, what with mass shipments of drives and studios and other companies switching loyalties.

On the other hand, some people are tired of the deluge of news coming in about HD-DVD/Blu-ray. It might be just as well that you sit back and wait for a 50 GB (should that be 45 GB?) drive to ship to a store near you—and then go upgrade! ■

ram_mohan@thinkdigit.com

Enhanced Security In Blu-ray

Copy-protection schemes on blue-laser discs are more advanced. BD+ and ROM-Mark are specific to Blu-ray; the scheme called AACS will be used by both HD-DVD and Blu-ray.

BD+ is an encryption scheme that allows for dynamically changing encryption schemes. If the encryption scheme is hacked, as happened in the DeCSS, it can be updated and used on all new discs.

Why is this required? Content-Scrambling System (CSS) is an unsuccessful encryption system used on some DVDs. When the contents of a DVD are encrypted, "keys" are required to unlock the content—more technically, "key sets" (because there is more than one key involved). These key sets are licensed to manufacturers, who incorporate them into DVD drives and movies. As a matter of fact, the CSS algorithm was reverse-engineered in 1999, and a program called DeCSS was released onto the Internet. And from then on, all DVDs became crackable! BD+ will avoid such a scenario.

ROM-Mark is a watermark that players need to see before the disc can be played. Only licensed parties will be able to produce the watermark, and when you copy a disc, the watermark, which is undetectable, will not be copied—so piracy is, in theory, defeated. In addition, if the watermark is detected by any means, the ROM-producing devices, into which the technology will be built, will not reproduce the content.